Disaster Aid — Analytics Report

Operational, volunteer and ML-model overview

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Key Performance Indicators

TOTAL REPORTS **125**	VERIFIED REPORTS **89 (71%)**	HELP REQUESTS **77**
RESOLVED REQUESTS **51 (66%)**	AVG RESPONSE TIME **22.0 min**	ACTIVE ML MODELS **2**
VOLUNTEER HOURS **247.5**	AVG CHURN PROBABILITY **53%**	HIGH-RISK VOLUNTEERS **2**

Executive Summary

This report consolidates the live state of the Disaster Aid platform. To date, the system has processed 125 citizen reports (71% verified) and routed 77 help requests with a 66% resolution rate. The mean response time across all incidents is 22.0 minutes.

On the volunteer side, 248 approved volunteer hours have been logged across 2 tracked organisations. The platform's churn predictor flags 2 volunteers as high risk (probability above 70%), with an overall average churn probability of 53%.

The ML stack currently runs 2 active models, covering disaster spread, volunteer churn and credibility scoring. Detailed breakdowns and recent activity follow on the next pages.

Disaster Analytics

Aggregated KPIs computed from **disaster_statistics** joined with **audit_log** and **predictive_model**.

Total citizen reports	**125**
Verified reports	**89**
Verification rate	**71%**
Total help requests	**77**
Resolved requests	**51**
Resolution rate	**66%**
Avg response time (min)	**22.00**
Audit log entries	**2**
Active predictive models	**2**

Volunteer Analytics

Aggregated KPIs computed from **volunteer_statistics**, **churn_prediction** and **organisation_stats**.

Total approved hours	**247.5**
Total opportunities	**66**
Average churn probability	**53%**
High-risk volunteers (> 70%)	**2**
Organisations tracked	**2**

Active Predictive Models

Name	Version	Type	Trained at
Credibility Score V2	2.1.0	CREDIBILITY_SCORER	2026-04-21 18:17
Spread V1	1.0.0	SPREAD_PREDICTOR	2026-04-15 18:17

Recent Audit Activity

Last entries from **audit_log**, ordered by timestamp.

No audit log entries recorded yet.

High-Risk Volunteer Churn

Top entries from **churn_prediction** with predicted probability above the 70% threshold.

Volunteer	Churn prob.	Intervention	Predicted at	Reactivated
2003	91%	ACHIEVEMENT_NUDGE	2026-04-27 18:07	no
2001	83%	TEAM_CHALLENGE	2026-04-27 17:22	no

Observations

- Only 66% of help requests are resolved — investigate backlog and responder availability.
- Average volunteer churn probability (53%) is at or above 50% — cohort retention is a risk.
- Top high-risk churn case sits at 91% predicted probability (2 volunteers above threshold).

Recommendations

- Trigger retention campaigns (achievement nudges, team challenges) for volunteers above the 40% churn threshold.
- Validate data freshness before publishing externally — ensure the disaster and volunteer statistics tables were re-computed within the last 24 hours.